

#This code will be used to demonstrate proposition 8.1. We demonstrate the existence of the three curves $\gamma 5$, $\gamma 6$ and $\gamma 7$ in region R2.

#Construct the polynomial g1 described in the text of section 8.1.

$$g1 := -(z^{-3} - t^{-3}) \cdot (z^{-3} - s^{-3}) + (x^{-3} - t^{-3}) \cdot (y^{-3} - s^{-3})$$

$$g1 := -\left(\frac{1}{z^3} - \frac{1}{t^3}\right) \left(\frac{1}{z^3} - \frac{1}{s^3}\right) + \left(\frac{1}{x^3} - \frac{1}{t^3}\right) \left(\frac{1}{y^3} - \frac{1}{s^3}\right) \quad (1)$$

#Describe s and t as a function of the other distances.

$$s := \frac{t \cdot y}{x} ;$$

$$t := \frac{x}{x + y} \cdot w ;$$

#Apply the rational parametrization constructed in this chapter 4..

$$\text{parametrization} := \left\{ w = a + \frac{1}{a}, \right.$$

$$z = b + \frac{1}{b},$$

$$y = a - \frac{1}{a} + b - \frac{1}{b},$$

$$x = a - \frac{1}{a} - b + \frac{1}{b} \left. \right\} :$$

simplify(numer(expand(subs(parametrization, g1)))) ;

denom(expand(subs(parametrization, g1))) ;

$$-\left(a^{18} b^6 + 3 (-b^8 + b^6 - b^4) a^{16} - 16 b^3 (b^2 + 1)^3 a^{15} + 3 (b^{10} - 3 b^8 + 4 b^6 - 3 b^4 + b^2) a^{14} - 48 b (b^4 - 4 b^2 + 1) (b^2 + 1)^3 a^{13} + (14 b^{12} + 99 b^{10} + 207 b^8 + 328 b^6 + 207 b^4 + 99 b^2 + 14) a^{12} + 192 (b^2 + 1)^3 \left(b^4 - \frac{13}{4} b^2 + 1 \right) b a^{11} + 3 (-18 b^{12} - 98 b^{10} - 265 b^8 - 328 b^6 - 265 b^4 - 98 b^2 - 18) a^{10} - 288 \left(b^4 - \frac{28}{9} b^2 + 1 \right) (b^2 + 1)^3 b a^9 + 3 (14 b^{12} + 94 b^{10} + 215 b^8 + 312 b^6 + 215 b^4 + 94 b^2 + 14) a^8 + 192 (b^2 + 1)^3 \left(b^4 - \frac{13}{4} b^2 + 1 \right) b a^7 + 3 (-6 b^{12} - 31 b^{10} - 91 b^8 - 104 b^6 - 91 b^4 - 31 b^2 - 6) a^6 - 48 b (b^4 - 4 b^2 + 1) (b^2 + 1)^3 a^5 + 3 (b^{10} - 3 b^8 + 4 b^6 - 3 b^4 + b^2) a^4 \right)$$

$$\begin{aligned}
& -16b^3(b^2+1)^3a^3 + 3(-b^8+b^6-b^4)a^2+b^6 \Big) a^3b^6 \\
& (a^2b-ab^2+a-b)^3(a^2b+ab^2-a-b)^3(b^2+1)^6(a^2+1)^3a^3
\end{aligned} \tag{2}$$

#Define r5

$$\begin{aligned}
r5 := & \left(a^{18}b^6 + 3(-b^8+b^6-b^4)a^{16} - 16b^3(b^2+1)^3a^{15} + 3(b^{10}-3b^8+4b^6-3b^4 \right. \\
& + b^2) a^{14} - 48b(b^4-4b^2+1)(b^2+1)^3a^{13} + (14b^{12}+99b^{10}+207b^8+328b^6+207b^4 \\
& + 99b^2+14)a^{12} + 192(b^2+1)^3 \left(b^4 - \frac{13}{4}b^2+1 \right) b a^{11} + 3(-18b^{12}-98b^{10}-265b^8 \\
& - 328b^6-265b^4-98b^2-18)a^{10} - 288 \left(b^4 - \frac{28}{9}b^2+1 \right) (b^2+1)^3 b a^9 + 3(14b^{12} \\
& + 94b^{10}+215b^8+312b^6+215b^4+94b^2+14)a^8 + 192(b^2+1)^3 \left(b^4 - \frac{13}{4}b^2+1 \right) b a^7 \\
& + 3(-6b^{12}-31b^{10}-91b^8-104b^6-91b^4-31b^2-6)a^6 - 48b(b^4-4b^2 \\
& + 1)(b^2+1)^3a^5 + 3(b^{10}-3b^8+4b^6-3b^4+b^2)a^4 - 16b^3(b^2+1)^3a^3 + 3(-b^8 \\
& + b^6-b^4)a^2+b^6 \Big) :
\end{aligned}$$

#Plot the curves r5=0 in region R2.

$r := a - b :$

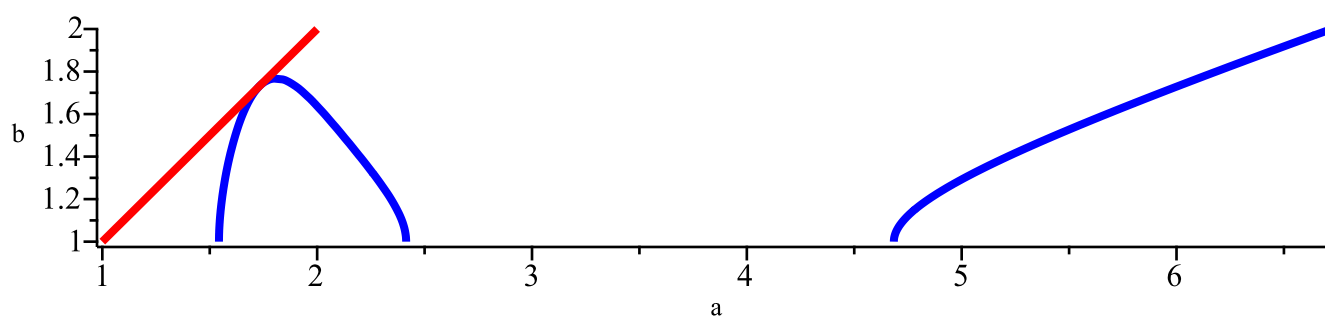
with(plots) :

```

implicitplot([ r5=0, r=0], a=1 .. 7, b=1 .. 2,
  scaling=constrained,
  gridrefine=2,
  thickness=3,
  color=[blue, red],
  labels=["a", "b"],
  title="Graphic for r5=0.");

```

Graphic for $r_5=0$.



#Apply the Möbius transformation on the rectangle $A1 = [1.5, \sqrt{3}) \times [1, \sqrt{3})$ constructing $r5$

$$\text{moebius} := \left\{ \begin{array}{l} a = \frac{\frac{3}{2} + \sqrt{3} \cdot c}{1 + c}, \\ b = \frac{1 + \sqrt{3} \cdot d}{1 + d} \end{array} \right\} :$$

$r5 := \text{collect}(\text{numer}(\text{expand}(\text{subs}(\text{moebius}, r5))), c) :$

#Compute the degree of $r5$ in c

$\text{degree}(r5, c);$

18

(3)

*#Define the coefficients $ci = \text{coeff}(r5, c, i)$
 $\text{seq}(\text{evalf}(\text{assign}(\text{cat}('c', i), \text{coeff}(r5, c, i))), i = 0 .. 18);$*

*#Compute the all degree of ci in d
 $\text{seq}(\text{degree}(\text{eval}(\text{cat}('c', i))), d), i = 0 .. 18);$*

12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 11

(4)

#From here we will analyze the signal variation of all coefficients ci

$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(c18, d, i))), i = 12 .. 0, -1);$

1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1

(5)

$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(c17, d, i))), i = 12 .. 0, -1);$

$\text{fsolve}(c17 = 0, d, 0 .. \text{infinity});$

-1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1

49.43281530

(6)

$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(c16, d, i))), i = 12 .. 0, -1);$

$\text{fsolve}(c16 = 0, d, 0 .. \text{infinity});$

-1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1

21.67161010

(7)

$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(c15, d, i))), i = 12 .. 0, -1);$

$\text{fsolve}(c15 = 0, d, 0 .. \text{infinity});$

-1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1

12.70412277

(8)

$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(c14, d, i))), i = 12 .. 0, -1);$

$$\begin{aligned}
 &fsolve(c14=0, d, 0 \dots \infty); \\
 &\quad -1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \\
 &\quad 8.384899311
 \end{aligned} \tag{9}$$

$$\begin{aligned}
 &seq(sign(evalf(coeff(c13, d, i))), i = 12 \dots 0, -1); \\
 &fsolve(c13=0, d, 0 \dots \infty); \\
 &\quad -1, -1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \\
 &\quad 5.895897942
 \end{aligned} \tag{10}$$

$$\begin{aligned}
 &seq(sign(evalf(coeff(c12, d, i))), i = 12 \dots 0, -1); \\
 &fsolve(c12=0, d, 0 \dots \infty); \\
 &\quad -1, -1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \\
 &\quad 4.303914833
 \end{aligned} \tag{11}$$

$$\begin{aligned}
 &seq(sign(evalf(coeff(c11, d, i))), i = 12 \dots 0, -1); \\
 &fsolve(c11=0, d, 0 \dots \infty); \\
 &\quad -1, -1, -1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \\
 &\quad 3.212580371
 \end{aligned} \tag{12}$$

$$\begin{aligned}
 &seq(sign(evalf(coeff(c10, d, i))), i = 12 \dots 0, -1); \\
 &fsolve(c10=0, d, 0 \dots \infty); \\
 &\quad -1, -1, -1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \\
 &\quad 2.425825868
 \end{aligned} \tag{13}$$

$$\begin{aligned}
 &seq(sign(evalf(coeff(c9, d, i))), i = 12 \dots 0, -1); \\
 &fsolve(c9=0, d, 0 \dots \infty); \\
 &\quad -1, -1, -1, -1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \\
 &\quad 1.835928720
 \end{aligned} \tag{14}$$

$$\begin{aligned}
 &seq(sign(evalf(coeff(c8, d, i))), i = 12 \dots 0, -1); \\
 &fsolve(c8=0, d, 0 \dots \infty); \\
 &\quad -1, -1, -1, -1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \\
 &\quad 1.378793528
 \end{aligned} \tag{15}$$

$$\begin{aligned}
 &seq(sign(evalf(coeff(c7, d, i))), i = 12 \dots 0, -1); \\
 &fsolve(c7=0, d, 0 \dots \infty); \\
 &\quad -1, -1, -1, -1, -1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \\
 &\quad 1.013518212
 \end{aligned} \tag{16}$$

$$\begin{aligned}
 &seq(sign(evalf(coeff(c6, d, i))), i = 12 \dots 0, -1); \\
 &fsolve(c6=0, d, 0 \dots \infty); \\
 &\quad -1, -1, -1, -1, -1, -1, 1, 1, 1, 1, 1, 1, 1, 1 \\
 &\quad 0.7117148453
 \end{aligned} \tag{17}$$

$$\begin{aligned}
 &seq(sign(evalf(coeff(c5, d, i))), i = 12 \dots 0, -1); \\
 &fsolve(c5=0, d, 0 \dots \infty); \\
 &\quad -1, -1, -1, -1, -1, -1, -1, -1, 1, 1, 1, 1, 1, 1 \\
 &\quad 0.4499687235
 \end{aligned} \tag{18}$$

$$\begin{aligned}
& seq(sign(evalf(coeff(c4, d, i))), i = 12 .. 0, -1); \\
& fsolve(c4 = 0, d, 0 .. infinity); \\
& \quad -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, 1, 1, 1 \\
& \quad 0.1930538225 \tag{19}
\end{aligned}$$

$$\begin{aligned}
& seq(sign(evalf(coeff(c3, d, i))), i = 12 .. 0, -1); \\
& \quad -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \tag{20}
\end{aligned}$$

$$\begin{aligned}
& seq(sign(evalf(coeff(c2, d, i))), i = 12 .. 0, -1); \\
& \quad -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \tag{21}
\end{aligned}$$

$$\begin{aligned}
& seq(sign(evalf(coeff(c1, d, i))), i = 12 .. 0, -1); \\
& \quad -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \tag{22}
\end{aligned}$$

$$\begin{aligned}
& seq(sign(evalf(coeff(c0, d, i))), i = 12 .. 0, -1); \\
& \quad -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \tag{23}
\end{aligned}$$

#This concludes the existence of the curve γ_5 in A1.

restart;

#Define r5

$$\begin{aligned}
 r5 := & \left(a^{18} b^6 + 3 (-b^8 + b^6 - b^4) a^{16} - 16 b^3 (b^2 + 1)^3 a^{15} + 3 (b^{10} - 3 b^8 + 4 b^6 - 3 b^4 \right. \\
 & + b^2) a^{14} - 48 b (b^4 - 4 b^2 + 1) (b^2 + 1)^3 a^{13} + (14 b^{12} + 99 b^{10} + 207 b^8 + 328 b^6 + 207 b^4 \\
 & + 99 b^2 + 14) a^{12} + 192 (b^2 + 1)^3 \left(b^4 - \frac{13}{4} b^2 + 1 \right) b a^{11} + 3 (-18 b^{12} - 98 b^{10} - 265 b^8 \\
 & - 328 b^6 - 265 b^4 - 98 b^2 - 18) a^{10} - 288 \left(b^4 - \frac{28}{9} b^2 + 1 \right) (b^2 + 1)^3 b a^9 + 3 (14 b^{12} \\
 & + 94 b^{10} + 215 b^8 + 312 b^6 + 215 b^4 + 94 b^2 + 14) a^8 + 192 (b^2 + 1)^3 \left(b^4 - \frac{13}{4} b^2 + 1 \right) b a^7 \\
 & + 3 (-6 b^{12} - 31 b^{10} - 91 b^8 - 104 b^6 - 91 b^4 - 31 b^2 - 6) a^6 - 48 b (b^4 - 4 b^2 \\
 & + 1) (b^2 + 1)^3 a^5 + 3 (b^{10} - 3 b^8 + 4 b^6 - 3 b^4 + b^2) a^4 - 16 b^3 (b^2 + 1)^3 a^3 + 3 (-b^8 \\
 & + b^6 - b^4) a^2 + b^6 \Big) :
 \end{aligned}$$

#Apply the Möbius transformation on the rectangle $A2 = [1.5, \text{sqrt}(3)) \times [1, \text{sqrt}(3))$ constructing r5

$$\begin{aligned}
 \text{moebius} := & \left\{ a = \frac{\text{sqrt}(3) + (1 + \text{sqrt}(2)) \cdot c}{1 + c}, \right. \\
 & \left. b = \frac{1 + 2 \cdot d}{1 + d} \right\} :
 \end{aligned}$$

r5 := collect(numer(expand(subs(moebius, r5))), d) :

#Compute the degree of r5 in d

degree(r5, d);

12

(24)

#Define the coefficients di = coeff(r5, d, i)

seq(evalf(assign(cat('d', i), coeff(r5, d, i))), i=0 .. 12);

#Compute the all degree of di in c

seq(degree(eval(cat('d', i)), c), i=0 .. 12);

17, 17, 18, 18, 18, 18, 18, 18, 18, 18, 18, 18

(25)

seq(sign(evalf(coeff(d12, c, i))), i=18 .. 0, -1);

-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1

(26)

seq(sign(evalf(coeff(d11, c, i))), i=18 .. 0, -1);

-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1

(27)

seq(sign(evalf(coeff(d10, c, i))), i=18 .. 0, -1);

fsolve(d10=0, c, 0 .. infinity);

$$\begin{aligned} & -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, 1, 1, 1, 1, -1 \\ & 0.05218070652, 0.2106357834 \end{aligned} \quad (28)$$

$$\begin{aligned} & seq(sign(evalf(coeff(d9, c, i))), i = 18 .. 0, -1); \\ & fsolve(d9 = 0, c, 0 .. infinity); \\ & -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, 1, 1, 1, 1, 1, 1, 1, 1 \\ & 0.5839602874 \end{aligned} \quad (29)$$

$$\begin{aligned} & seq(sign(evalf(coeff(d8, c, i))), i = 18 .. 0, -1); \\ & fsolve(d8 = 0, c, 0 .. infinity); \\ & -1, -1, -1, -1, -1, -1, -1, -1, -1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \\ & 0.9850809934 \end{aligned} \quad (30)$$

$$\begin{aligned} & seq(sign(evalf(coeff(d7, c, i))), i = 18 .. 0, -1); \\ & fsolve(d7 = 0, c, 0 .. infinity); \\ & -1, -1, -1, -1, -1, -1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \\ & 1.546567948 \end{aligned} \quad (31)$$

$$\begin{aligned} & seq(sign(evalf(coeff(d6, c, i))), i = 18 .. 0, -1); \\ & fsolve(d6 = 0, c, 0 .. infinity); \\ & -1, -1, -1, -1, -1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \\ & 2.419333327 \end{aligned} \quad (32)$$

$$\begin{aligned} & seq(sign(evalf(coeff(d5, c, i))), i = 18 .. 0, -1); \\ & fsolve(d5 = 0, c, 0 .. infinity); \\ & -1, -1, -1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \\ & 3.926440744 \end{aligned} \quad (33)$$

$$\begin{aligned} & seq(sign(evalf(coeff(d4, c, i))), i = 18 .. 0, -1); \\ & fsolve(d4 = 0, c, 0 .. infinity); \\ & -1, -1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \\ & 6.923819135 \end{aligned} \quad (34)$$

$$\begin{aligned} & seq(sign(evalf(coeff(d3, c, i))), i = 18 .. 0, -1); \\ & fsolve(d3 = 0, c, 0 .. infinity); \\ & -1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \\ & 14.40787537 \end{aligned} \quad (35)$$

$$\begin{aligned} & seq(sign(evalf(coeff(d2, c, i))), i = 18 .. 0, -1); \\ & fsolve(d2 = 0, c, 0 .. infinity); \\ & -1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \\ & 44.37861823 \end{aligned} \quad (36)$$

$$\begin{aligned} & seq(sign(evalf(coeff(d1, c, i))), i = 18 .. 0, -1); \\ & 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \end{aligned} \quad (37)$$

$$\begin{aligned} & seq(sign(evalf(coeff(d0, c, i))), i = 18 .. 0, -1); \\ & 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \end{aligned} \quad (38)$$

#This concludes the existence of the curve γ_6 in A2.

restart;

#Define r5

$$\begin{aligned}
 r5 := & \left(a^{18} b^6 + 3 (-b^8 + b^6 - b^4) a^{16} - 16 b^3 (b^2 + 1)^3 a^{15} + 3 (b^{10} - 3 b^8 + 4 b^6 - 3 b^4 \right. \\
 & + b^2) a^{14} - 48 b (b^4 - 4 b^2 + 1) (b^2 + 1)^3 a^{13} + (14 b^{12} + 99 b^{10} + 207 b^8 + 328 b^6 + 207 b^4 \\
 & + 99 b^2 + 14) a^{12} + 192 (b^2 + 1)^3 \left(b^4 - \frac{13}{4} b^2 + 1 \right) b a^{11} + 3 (-18 b^{12} - 98 b^{10} - 265 b^8 \\
 & - 328 b^6 - 265 b^4 - 98 b^2 - 18) a^{10} - 288 \left(b^4 - \frac{28}{9} b^2 + 1 \right) (b^2 + 1)^3 b a^9 + 3 (14 b^{12} \\
 & + 94 b^{10} + 215 b^8 + 312 b^6 + 215 b^4 + 94 b^2 + 14) a^8 + 192 (b^2 + 1)^3 \left(b^4 - \frac{13}{4} b^2 + 1 \right) b a^7 \\
 & + 3 (-6 b^{12} - 31 b^{10} - 91 b^8 - 104 b^6 - 91 b^4 - 31 b^2 - 6) a^6 - 48 b (b^4 - 4 b^2 \\
 & + 1) (b^2 + 1)^3 a^5 + 3 (b^{10} - 3 b^8 + 4 b^6 - 3 b^4 + b^2) a^4 - 16 b^3 (b^2 + 1)^3 a^3 + 3 (-b^8 \\
 & + b^6 - b^4) a^2 + b^6) :
 \end{aligned}$$

#Apply the Möbius transformation on the rectangle $A3 = [1, 1.5) \times [1, 1.5)$ constructing r5

$$\begin{aligned}
 \text{moebius} := & \left\{ a = \frac{1 + \frac{3}{2} \cdot c}{1 + c}, \right. \\
 b = & \left. \frac{1 + \frac{3}{2} \cdot d}{1 + d} \right\} :
 \end{aligned}$$

r5 := collect(numer(expand(subs(moebius, r5))), c) :

#Compute the degree of r5 in c

degree(r5, c);

18

(39)

#Define the coefficients ci = coeff(r5, c, i)

for i from 0 to 18 do

 c[i] := coeff(r5, c, i);

od;

#Compute the all degree of ci in d

seq(degree(c[i], d), i = 0..18);

12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12

(40)

#From here we will analyze the signal variation of all coefficients c_i

$$\begin{aligned} & seq(seq(sign(coeff(c[j], d, i)), i = 12..0, -1), j = 18..0, -1); \\ & -1, \quad (41) \\ & -1, \\ & -1, \\ & -1, \\ & -1, \\ & -1, \\ & -1, \\ & -1, \\ & -1, \\ & -1, \\ & -1, \\ & -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \end{aligned}$$

#This concludes that $r_5=0$ does not intersect the rectangle $A_3=[1, 1.5) \times [1, 1.5)$.

restart;

#Define r5

$$\begin{aligned} r5 := & \left(a^{18} b^6 + 3 (-b^8 + b^6 - b^4) a^{16} - 16 b^3 (b^2 + 1)^3 a^{15} + 3 (b^{10} - 3 b^8 + 4 b^6 - 3 b^4 \right. \\ & + b^2) a^{14} - 48 b (b^4 - 4 b^2 + 1) (b^2 + 1)^3 a^{13} + (14 b^{12} + 99 b^{10} + 207 b^8 + 328 b^6 + 207 b^4 \\ & + 99 b^2 + 14) a^{12} + 192 (b^2 + 1)^3 \left(b^4 - \frac{13}{4} b^2 + 1 \right) b a^{11} + 3 (-18 b^{12} - 98 b^{10} - 265 b^8 \\ & - 328 b^6 - 265 b^4 - 98 b^2 - 18) a^{10} - 288 \left(b^4 - \frac{28}{9} b^2 + 1 \right) (b^2 + 1)^3 b a^9 + 3 (14 b^{12} \\ & + 94 b^{10} + 215 b^8 + 312 b^6 + 215 b^4 + 94 b^2 + 14) a^8 + 192 (b^2 + 1)^3 \left(b^4 - \frac{13}{4} b^2 + 1 \right) b a^7 \\ & + 3 (-6 b^{12} - 31 b^{10} - 91 b^8 - 104 b^6 - 91 b^4 - 31 b^2 - 6) a^6 - 48 b (b^4 - 4 b^2 \\ & + 1) (b^2 + 1)^3 a^5 + 3 (b^{10} - 3 b^8 + 4 b^6 - 3 b^4 + b^2) a^4 - 16 b^3 (b^2 + 1)^3 a^3 + 3 (-b^8 \\ & + b^6 - b^4) a^2 + b^6) : \end{aligned}$$

#Apply the Möbius transformation on the rectangle A4 = [1+sqrt(2), 4) × [1, 4) constructing r5

$$\begin{aligned} moebius := & \left\{ a = \frac{1 + \sqrt{2} + 4 \cdot c}{1 + c}, \right. \\ & \left. b = \frac{1 + 4 \cdot d}{1 + d} \right\} : \end{aligned}$$

r5 := collect(numer(expand(subs(moebius, r5))), c) :

#Compute the degree of r5 in c

degree(r5, c);

18

(42)

#Define the coefficients ci = coeff(r5, c, i)

for i from 0 to 18 do

c[i] := coeff(r5, c, i);

od;

#Compute the all degree of ci in d

seq(degree(c[i], d), i=0..18);

12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12

(43)

#From here we will analyze the signal variation of all coefficients ci

[illegible]

#This concludes that $r_5=0$ does not intersect the rectangle $A_4=[1 + \sqrt{2}, 4) \times [1, 4)$.

```
#Define r5
```

#Apply the Möbius transformation on the infinte trapezoid $A5 = \{(a,b); 4 < a, 1 < b < a\}$ constructing $r5$

```
r5 := collect(numer(expand(subs(moebius, r5))), c) :
```

$$degree(r_5, c);$$

(46)

[illegible]
$$seq(sign(evalf(coeff(c24, d, i))), i = 12 .. 0, -1);$$

$$\begin{aligned}
& c24; \\
& fsolve(c24=0, d, 0 \dots \infty); \\
& \quad -1, -1, -1, -1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \\
& \quad -49 d^{12} - 96 d^{11} - 48 d^{10} - 8 d^9 + 12 d^8 + 6 d^7 + d^6 \\
& \quad 0., 0., 0., 0., 0., 0., 0.4838711307
\end{aligned} \tag{48}$$

$$\begin{aligned}
& seq(sign(evalf(coeff(c23, d, i))), i = 12 \dots 0, -1); \\
& c23; \\
& fsolve(c23=0, d, 0 \dots \infty); \\
& \quad -1, -1, -1, -1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \\
& \quad -4704 d^{12} - 9708 d^{11} - 5568 d^{10} - 1224 d^9 + 1032 d^8 + 642 d^7 + 132 d^6 + 6 d^5 \\
& \quad 0., 0., 0., 0., 0., 0.4705128005
\end{aligned} \tag{49}$$

$$\begin{aligned}
& seq(sign(evalf(coeff(c22, d, i))), i = 12 \dots 0, -1); \\
& fsolve(c22=0, d, 0 \dots \infty); \\
& \quad -1, -1, -1, -1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \\
& \quad 0., 0., 0., 0., 0.4566315943
\end{aligned} \tag{50}$$

$$\begin{aligned}
& seq(sign(evalf(coeff(c21, d, i))), i = 12 \dots 0, -1); \\
& fsolve(c21=0, d, 0 \dots \infty); \\
& \quad -1, -1, -1, -1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \\
& \quad 0., 0., 0., 0.4422270787
\end{aligned} \tag{51}$$

$$\begin{aligned}
& seq(sign(evalf(coeff(c20, d, i))), i = 12 \dots 0, -1); \\
& fsolve(c20=0, d, 0 \dots \infty); \\
& \quad -1, -1, -1, -1, 1, 1, 1, 1, 1, 1, 1, 1, 1 \\
& \quad 0., 0., 0.4272958335
\end{aligned} \tag{52}$$

$$\begin{aligned}
& seq(sign(evalf(coeff(c19, d, i))), i = 12 \dots 0, -1); \\
& fsolve(c19=0, d, 0 \dots \infty); \\
& \quad -1, -1, -1, -1, -1, 1, 1, 1, 1, 1, 1, 1, 1 \\
& \quad 0., 0.4118312516
\end{aligned} \tag{53}$$

$$\begin{aligned}
& seq(sign(evalf(coeff(c18, d, i))), i = 12 \dots 0, -1); \\
& fsolve(c18=0, d, 0 \dots \infty); \\
& \quad -1, -1, -1, -1, -1, 1, 1, 1, 1, 1, 1, 1, 1 \\
& \quad 0.3958232316
\end{aligned} \tag{54}$$

$$\begin{aligned}
& seq(sign(evalf(coeff(c17, d, i))), i = 12 \dots 0, -1); \\
& fsolve(c17=0, d, 0 \dots \infty); \\
& \quad -1, -1, -1, -1, -1, 1, 1, 1, 1, 1, 1, 1, 1 \\
& \quad 0.3792577328
\end{aligned} \tag{55}$$

$$\begin{aligned}
& seq(sign(evalf(coeff(c16, d, i))), i = 12 \dots 0, -1); \\
& fsolve(c16=0, d, 0 \dots \infty); \\
& \quad -1, -1, -1, -1, -1, 1, 1, 1, 1, 1, 1, 1, 1 \\
& \quad 0.3621161449
\end{aligned} \tag{56}$$

$$\begin{aligned}
& seq(sign(evalf(coeff(c15, d, i))), i = 12 .. 0, -1); \\
& fsolve(c15 = 0, d, 0 .. infinity); \\
& \quad -1, -1, -1, -1, -1, 1, 1, 1, 1, 1, 1, 1, 1 \\
& \quad 0.3443743927
\end{aligned} \tag{57}$$

$$\begin{aligned}
& seq(sign(evalf(coeff(c14, d, i))), i = 12 .. 0, -1); \\
& fsolve(c14 = 0, d, 0 .. infinity); \\
& \quad -1, -1, -1, -1, -1, -1, 1, 1, 1, 1, 1, 1, 1 \\
& \quad 0.3260016547
\end{aligned} \tag{58}$$

$$\begin{aligned}
& seq(sign(evalf(coeff(c13, d, i))), i = 12 .. 0, -1); \\
& fsolve(c13 = 0, d, 0 .. infinity); \\
& \quad -1, -1, -1, -1, -1, -1, 1, 1, 1, 1, 1, 1, 1 \\
& \quad 0.3069584943
\end{aligned} \tag{59}$$

$$\begin{aligned}
& seq(sign(evalf(coeff(c12, d, i))), i = 12 .. 0, -1); \\
& fsolve(c12 = 0, d, 0 .. infinity); \\
& \quad -1, -1, -1, -1, -1, -1, 1, 1, 1, 1, 1, 1, 1 \\
& \quad 0.2871940669
\end{aligned} \tag{60}$$

$$\begin{aligned}
& seq(sign(evalf(coeff(c11, d, i))), i = 12 .. 0, -1); \\
& fsolve(c11 = 0, d, 0 .. infinity); \\
& \quad -1, -1, -1, -1, -1, -1, 1, 1, 1, 1, 1, 1, 1 \\
& \quad 0.2666418178
\end{aligned} \tag{61}$$

$$\begin{aligned}
& seq(sign(evalf(coeff(c10, d, i))), i = 12 .. 0, -1); \\
& fsolve(c10 = 0, d, 0 .. infinity); \\
& \quad -1, -1, -1, -1, -1, -1, -1, 1, 1, 1, 1, 1, 1 \\
& \quad 0.2452125986
\end{aligned} \tag{62}$$

$$\begin{aligned}
& seq(sign(evalf(coeff(c9, d, i))), i = 12 .. 0, -1); \\
& fsolve(c9 = 0, d, 0 .. infinity); \\
& \quad -1, -1, -1, -1, -1, -1, -1, 1, 1, 1, 1, 1, 1 \\
& \quad 0.2227831276
\end{aligned} \tag{63}$$

$$\begin{aligned}
& seq(sign(evalf(coeff(c8, d, i))), i = 12 .. 0, -1); \\
& fsolve(c8 = 0, d, 0 .. infinity); \\
& \quad -1, -1, -1, -1, -1, -1, -1, -1, 1, 1, 1, 1, 1 \\
& \quad 0.1991754467
\end{aligned} \tag{64}$$

$$\begin{aligned}
& seq(sign(evalf(coeff(c7, d, i))), i = 12 .. 0, -1); \\
& fsolve(c7 = 0, d, 0 .. infinity); \\
& \quad -1, -1, -1, -1, -1, -1, -1, -1, 1, 1, 1, 1, 1 \\
& \quad 0.1741173416
\end{aligned} \tag{65}$$

$$\begin{aligned}
& seq(sign(evalf(coeff(c6, d, i))), i = 12 .. 0, -1); \\
& fsolve(c6 = 0, d, 0 .. infinity); \\
& \quad -1, -1, -1, -1, -1, -1, -1, -1, 1, 1, 1, 1, 1
\end{aligned}$$

$$0.1471572226 \quad (66)$$

$$\begin{aligned} &seq(sign(evalf(coeff(c5, d, i))), i = 12 .. 0, -1); \\ &fsolve(c5 = 0, d, 0 .. infinity); \\ &\quad -1, -1, -1, -1, -1, -1, -1, -1, -1, 1, 1, 1, 1 \end{aligned}$$

$$0.1174485613 \quad (67)$$

$$\begin{aligned} &seq(sign(evalf(coeff(c4, d, i))), i = 12 .. 0, -1); \\ &fsolve(c4 = 0, d, 0 .. infinity); \\ &\quad -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, 1, 1, 1 \end{aligned}$$

$$0.08303077385 \quad (68)$$

$$\begin{aligned} &seq(sign(evalf(coeff(c3, d, i))), i = 12 .. 0, -1); \\ &fsolve(c3 = 0, d, 0 .. infinity); \\ &\quad -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, 1, 1 \end{aligned}$$

$$0.03626703711 \quad (69)$$

$$\begin{aligned} &seq(sign(evalf(coeff(c2, d, i))), i = 12 .. 0, -1); \\ &fsolve(c2 = 0, d, 0 .. infinity); \\ &\quad -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \end{aligned} \quad (70)$$

$$\begin{aligned} &seq(sign(evalf(coeff(c1, d, i))), i = 12 .. 0, -1); \\ &\quad -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \end{aligned} \quad (71)$$

$$\begin{aligned} &seq(sign(evalf(coeff(c0, d, i))), i = 12 .. 0, -1); \\ &\quad -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1 \end{aligned} \quad (72)$$

#This concludes the existence of the curve γ_7 in A5.